

Post-detection related developments, ongoing work towards SETI Protocols 2.0, and SETI in Society news.

An update for the IAA SETI Permanent Committee, 23 Oct 2019, at IAC Washington

Quick overview

- News. Recent updates from the community re: SETI/Society? (Claire?)
- Then detailed comments from Les and Kathryn and Additional Members re:
 - Status of Protocols and Post-Detection
 - Suggested next steps.

Post-det / SETI & Society News

- US National Academies Decadal Survey on Astronomy and Astrophysics (Astro2020) white papers: in addition to SETI science papers:
 - “Preparing for the Discovery of Life Beyond Earth”, Kathryn Denning and Steve Dick argued that the National Academies should “created a federal funding structure -- e.g., a NASA collaboration with NSF(SBE) and/or the National Endowment for the Humanities -- to foster sustained collaboration between scientists, social scientists, and the humanities addressing ongoing, emerging, and future societal issues resulting from intentional searches or serendipitous discoveries of extraterrestrial life.” 119 cosigners/endorsers from the global SETI and astrobiology community in the space of 24 hours, with additional cosigners sending their support after the deadline. (Thx SETI lists!)
 - Berea et al “Social Sciences Interdisciplinarity for Astronomy and Astrophysics -- Lessons from the History of NASA and Related Fields”
- Tennen & Forgan efforts over past two years
- Denning, Anamaria Berea, Klara Anna Capova, Duncan Forgan, Eric Korpela, Michael Oman-Reagan, Jill Tarter paper in SETI & Society session yesterday: “SETI and Post-detection: towards a new research roadmap”
- NASA Technosignatures Workshop Report (Dec 2018) included post-detection elements
- UK Royal Society summer 2019 outreach and survey
- AbSciCon sessions (Denning & Berea session, Haqq-Misra session, and public session L. Walkowicz et al)
- Green Bank workshop summer 2019 – S. Dick and others did papers on discovery/impact related themes
- Rio 2.0 publications late 2018
- Multiple papers within the SETI community re: METI, etc
- And lots more – individual papers, books, etc.

Protocols and Post-Detection... reloading *(watch this space)*

Les Tennen, Kathryn Denning

With inspiration / cumulative contributions from the other co-authors of Denning et al 2019 (Anamaria Berea, Klara Anna Capova, Duncan Forgan, Eric Korpela, Michael Oman-Reagan, Jill Tarter), Steve Dick, Ivan Almar, Michael Michaud, John Elliott, Lori Walton, Denise Herzing, Paul Davies, Milan Cirkovic, and *many more* on this committee and beyond who have done work in this area since SETI began (a tip of the hat to you).

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Composition of IAA SETI Committee in mid-2018 (~80)

<https://iaaseti.org/en/members/>

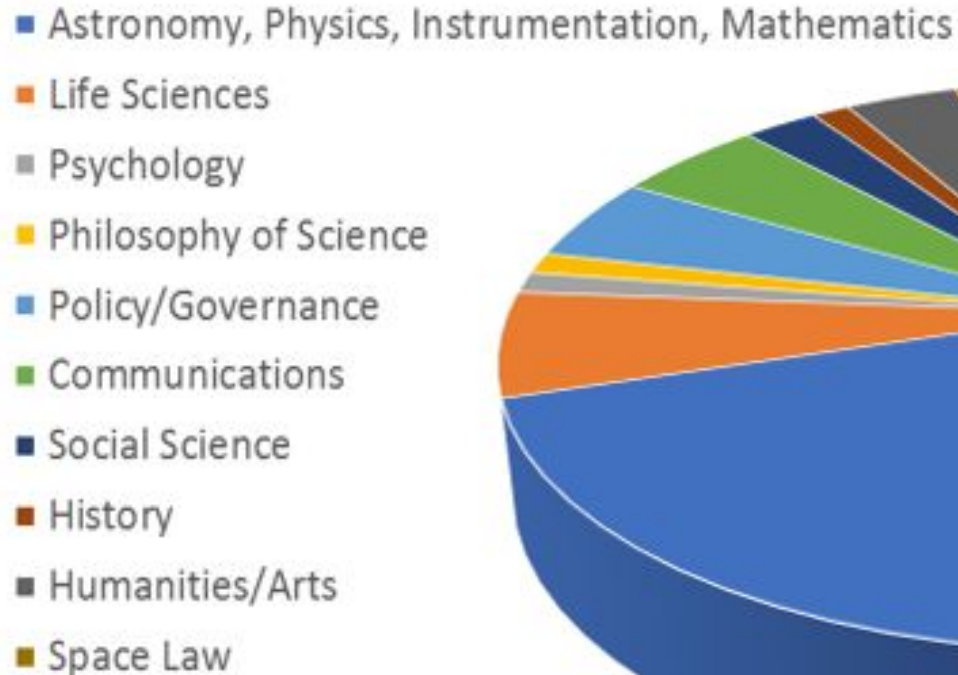
For reference:

Membership in this committee has fluctuated over the years, but it rose sharply several years ago.

- In 2002, 37 members (~ half had expertise other than astronomy/physics)
- In 2009, 43 members
- In 2014, 40 members
- In 2019, over 80 members

Also, as of mid-2018:

- female proportion of committee is < 20%
 - ~ 50% of committee is American
- A survey would tell us more (age etc)



Several suggestions flow from that reality of committee composition:

- Recognize that we do not have all the necessary expertise in-house for social & communications topics – we must draw on external expertise and research, but we should also bring more diverse experts in
- Hold a series of virtual meetings which are small enough that people can actually talk to each other
- Gather some information from committee members about interests/skills
- We need to recognize as well that some members cannot travel to IAC meetings and can only ever participate remotely.
- Given the attempt to integrate SETI/technosignatures and astrobiology, i.e. technosignatures are biosignatures... perhaps we need more astrobio.

Protocols and Post-Detection... reloading

The Status:

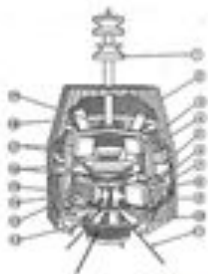
- 2010 Principles (revision of 1989) are already outdated
 - Post-detection Committee (*which is a part of those Principles*) hasn't existed for a few years and still needs to be reimagined
 - Rio (*the use of Rio or something like it, is part of the Principles*) has been improved into Rio 2.0 with significant strengths, but can be further adjusted, and/or a complementary strategy for public communications could be developed
- The Principles themselves may need an update but as Tennen and Forgan concluded in their papers over the last two years, some social research would help us figure out the path. But this isn't easy.
- This is one reason why Denning et al 2019 (Berea, Capova, Forgan, Korpela, Oman-Reagan, Tarter) suggested research roadmapping to figure out *what* we need to know, and how we can get that information. Thanks to Bill Diamond/SETI Inst for agreement in principle to move forward with this. (details coming)
- Many of the big challenges stem from huge changes in the communication landscape.

Evolution of communications systems since modern SETI began

1960s



1970s



1980s



1990s



2000s



2010s



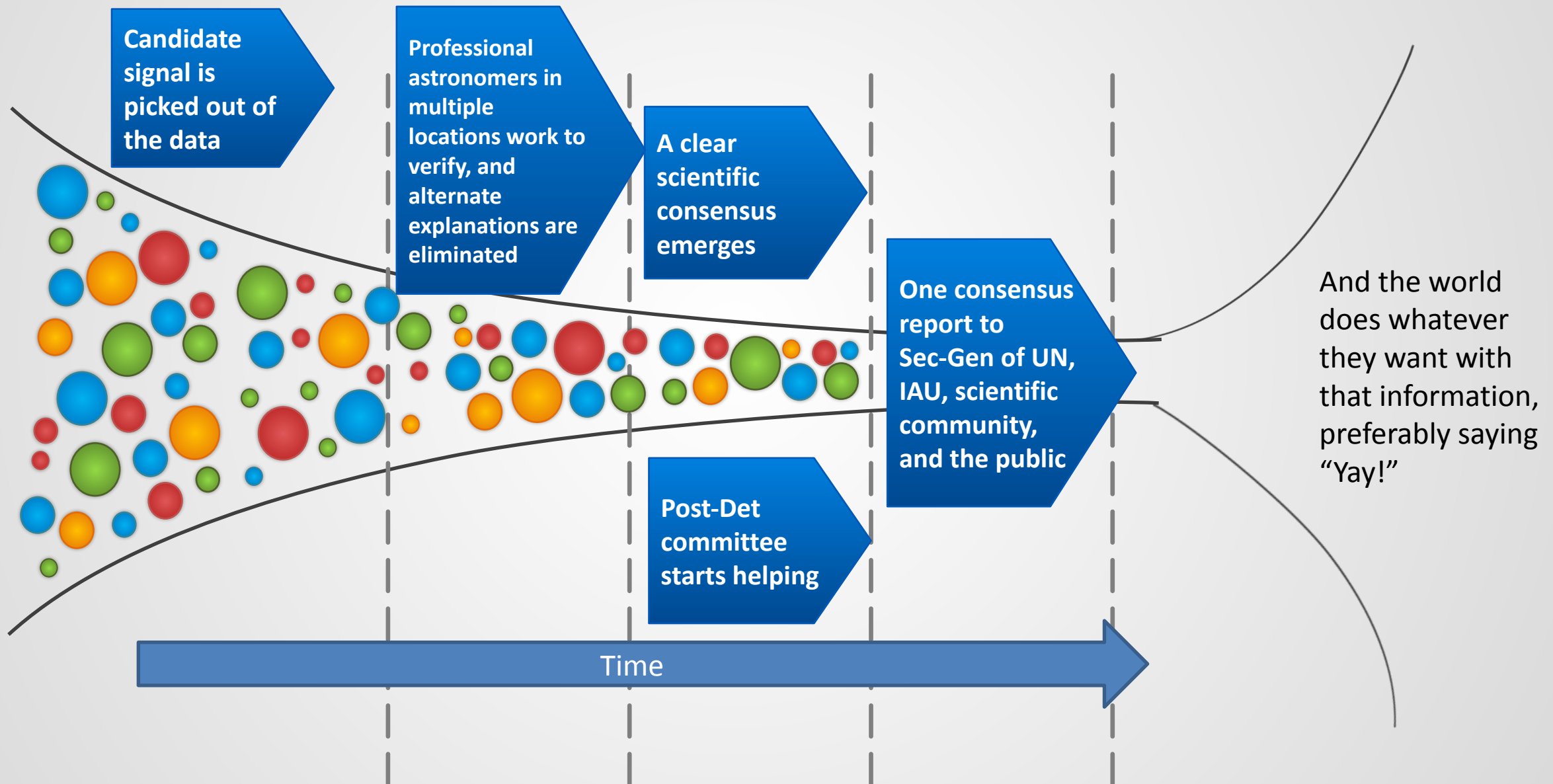
2020s



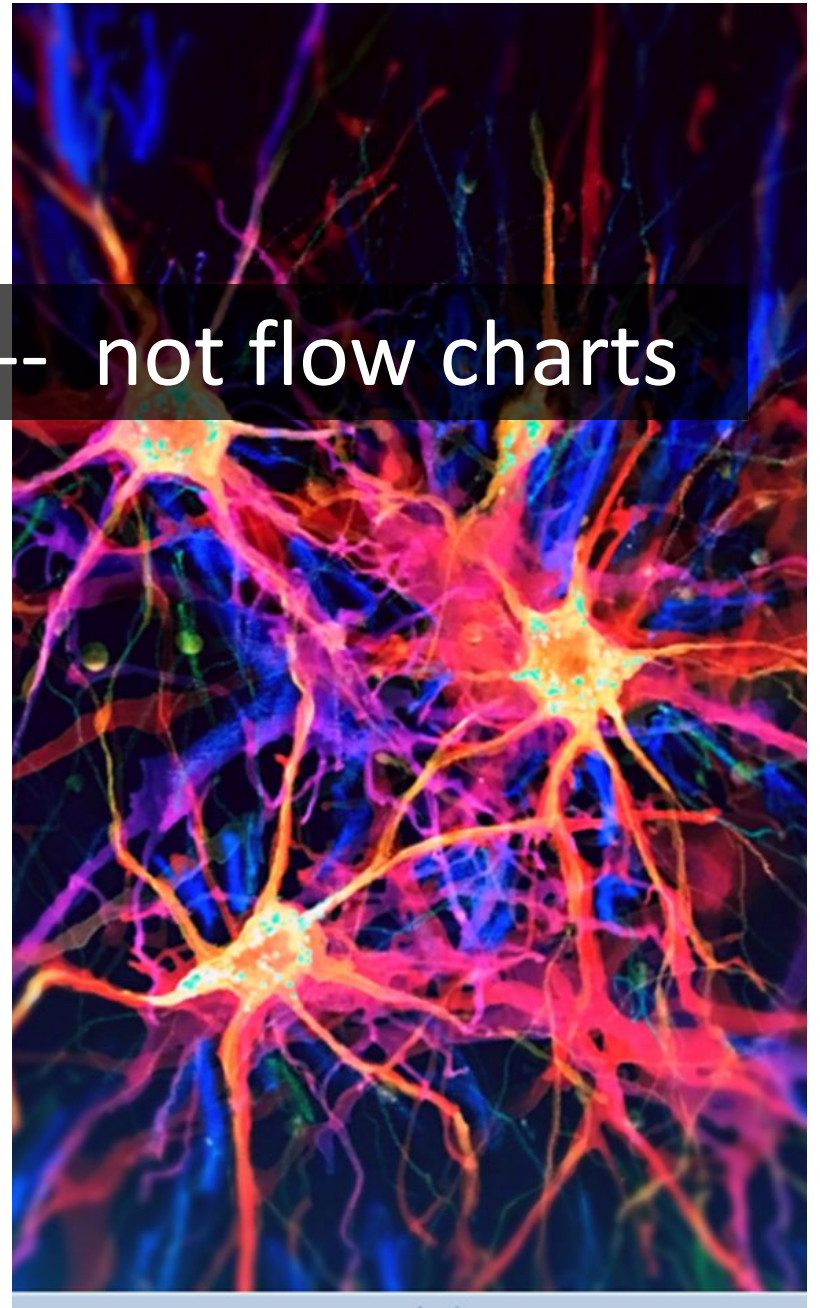
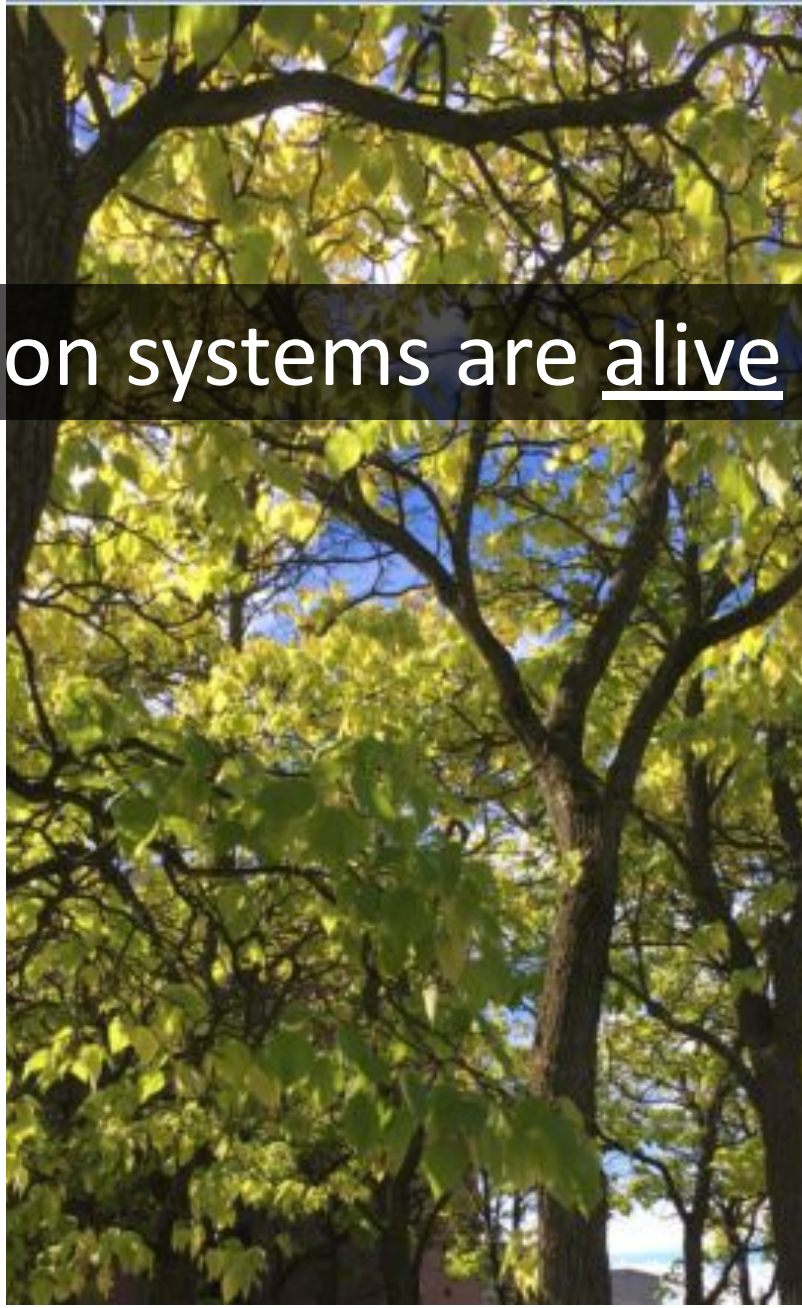
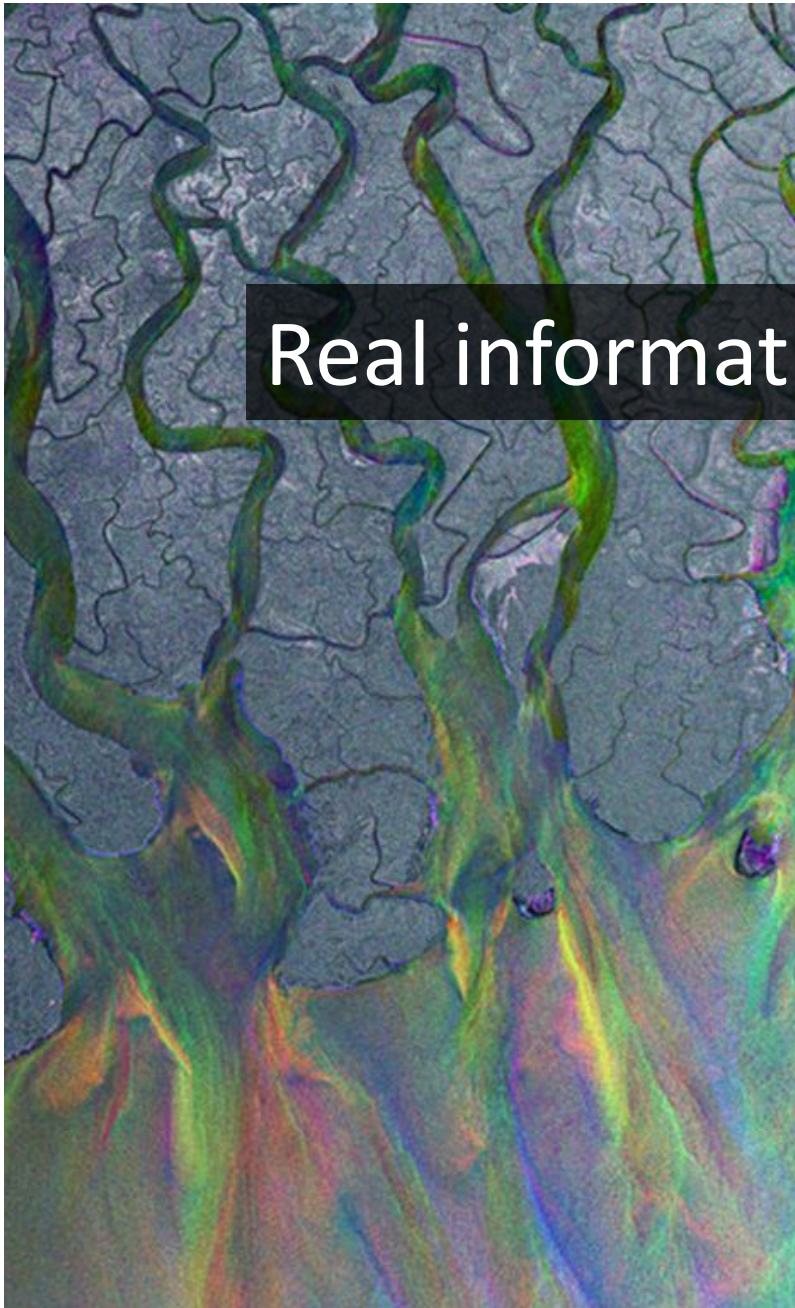
**Now: peer-to-peer, unmediated, fast, and wide:
Increase in direct communication between scientists
and scientists, scientists and public, faster
information transmission, etc...
Has implications for how we handle
announcements, debates, discussions**



Idealized process implicit in original Principles



Real information systems are alive -- not flow charts



Approaches and bodies of research that can help (there are more) (from Denning et al 2019)

- Information flow studies (information seeking behavior, including radio, TV, posters, word of mouth, text)
- Social media studies (including negativity bias, mis/disinformation, profit)
- Effective brief communication (e.g. emergency warning systems)
- Not just words: also images
- Mixed methods: interviews, focus groups, analogy, careful survey design
- Legal and Economic questions

so.... how do we move forward?

ultimately, by going back to the spirit of the Principles, rather than the letter

- Original intentions (approximately):
 - encourage researchers in different countries to share, by *mutually* promising to be forthcoming
 - to do responsible government-sponsored science
 - be transparent with the public
 - supply reliable, verified information to the world
 - be worthy of trust
 - Assure everyone that SETI researchers would not exceed scientific remit, i.e. would stop short of exopolitics, i.e. let world governments step in, if contact actually is made

And now we must ask:

- what does scientific responsibility look like *now*?
- what are the situations to be avoided now?
- what would a discovery look like *now*?
- what is in our hands, and what is out of our hands, *now*?
- how have our publics changed?

Suggested directions (these interconnect)

Aim for Principles AND Codes of Conduct AND Best Practices

Timeless principles

Voluntary
codes/agreements
among all levels of
research (incl
students, citizens)

Regularly updated,
research-based best
practices
suggestions

Research roadmap with broad scope

Social / communication
research topics such as these
above,

AND

Ethics

Studies of evolution of
intelligence &
communication

Studies of language /
decipherment

Urgent priority areas include:

Concerns of involved
scientists

Information seeking and
sharing behavior

Areas of public concern

Best practices for social
media use for
researchers

So, recommendations:

- 1) An informal, small interdisciplinary Committee working group for one year, which will rethink and recommend which subcomms we will need.
- 2) Hold a series of small Zoom meetings for Committee members to find out interests, expertises, and skills
- 3) Small groups to produce some draft Codes of Conduct – to be brought back to this committee next year
- 4) Small groups to produce some draft Best Practices – e.g. Social Media Safety – to be brought back to this committee within a year
- 5) Move ahead with Research Roadmapping for social research related to technosig discovery process and public communications. (KD will begin that process, with SI.)
- 6) And?